

TEJASWINI GUNNAPANENI

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PROFESSIONAL SUMMARY

Data Scientist and ML Engineer with 5 years of experience building production machine learning and Generative AI systems at scale. Skilled in developing forecasting models, multimodal learning systems, and LLM-powered assistants, with expertise across the full ML lifecycle from data engineering and modeling to evaluation, deployment, and monitoring. Proficient in Python, PyTorch/TensorFlow, cloud platforms (GCP/AWS), and implementing reliability best practices for enterprise AI applications.

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, ARIMA, CNNs, Transformers, time-series forecasting, anomaly detection, feature engineering, hyperparameter tuning, model explainability

Generative AI & LLMs: OpenAI APIs, Llama-based models, RAG pipelines, embedding/vector search, prompt engineering, tool calling, guardrails, evaluation harnesses, hallucination/error analysis

AI Development Tools: Cursor IDE, OpenAI Codex

Data & Platforms: Python, SQL, Pandas, NumPy, PySpark, BigQuery, SAP HANA/IBP (OData/SQL), PostgreSQL, Athena, S3 data lakes

MLOps & Deployment: DataRobot, Docker, Kubernetes, GitLab CI/CD, ArgoCD, Tekton, Terraform, Pulumi, experiment tracking, model validation, batch/API inference

Cloud & Observability: GCP (Vertex AI, BigQuery, Pub/Sub), AWS (S3, ECS, SageMaker basics, CloudWatch), Datadog, alerting, reliability dashboards

Software Engineering: FastAPI, Go microservices, REST APIs, async workflows (Celery/Redis), Jenkins, SonarQube, unit/integration testing, production debugging

Languages: Python, C++, SQL, R, Go, Java, CUDA

PROFESSIONAL EXPERIENCE

Nvidia — Machine Learning Engineer

May 2025 - Present

Built forecasting and GenAI systems for supply chain and enterprise analytics, owned data pipelines, model evaluation, and production deployment.

- Built a 28-week GPU demand forecasting pipeline; benchmarked XGBoost, LightGBM, CNN, and ARIMA on 400+ macro/supply features, improving forecast accuracy by 10-15%.
- Designed an LLM-powered analytics code-generation system using OpenAI APIs and Llama-based models; curated prompt-code-result datasets and evaluation dashboards; adopted by 50+ global planners.
- Developed an enterprise metrics assistant integrating SAP HANA/IBP data (OData/SQL) with retrieval and web search, added guardrails to block out-of-scope queries and improve answer reliability.
- Productionized services on NVIDIA internal ML platform (Astra via Pulumi) with Docker, GitLab CI/CD, and ArgoCD, monitored with Datadog for latency, errors, and uptime.
- Built reproducible data ingestion pipelines with AWS S3 for curated datasets and model artifacts, enabling consistent training, evaluation, and reporting across teams.

Delta Rising Foundation — Full Stack AI Engineer

February 2025 - April 2025

Built a multimodal deep learning pipeline for environmental monitoring and automated report generation for researchers.

- Developed a multimodal ML pipeline combining LiDAR and hyperspectral imagery with deep learning for forest monitoring and species classification across 25+ sites, achieving approximately 87% accuracy.
- Integrated transformer models for automatic ecological report generation, reducing manual documentation time by approximately 40%.

University of Central Missouri — Graduate Researcher

January 2024 - December 2024

Conducted research and built production-ready ML systems spanning speech recognition, multimodal GenAI applications, recommendation engines, and LLM-powered automation on cloud infrastructure.

- Conducted research on deep learning architectures for multilingual and multi-domain Automatic Speech Recognition (ASR), comparing model performance across languages and domains.
- Developed multimodal video analysis application with LLM-powered summarization and Q&A, improving content accessibility across text, image, and audio.
- Built and optimized RAG pipelines using ChromaDB and Pinecone, enhancing retrieval accuracy for question-answering systems.
- Implemented LLM automation tools with LangChain for text processing, improving efficiency by 25%.
- Developed song recommendation engine using k-NN and clustering on user behavior and audio embeddings (Databricks, Spotify API, MLlib).
- Deployed ML applications on AWS cloud infrastructure for high-performance AI workloads.

Ford Motor Company — Software Engineer - Data Scientist

September 2021 - May 2023

Built and deployed ML-based cloud security detection and remediation systems on GCP with scalable data processing and operational dashboards.

- Developed ML models for cloud security detection and deployed on GCP Vertex AI, improving security scores by 30-35% and reducing high-risk misconfigurations by approximately 25%.
- Built Golang remediation services and alerting systems (Pub/Sub, Webex, SendGrid), cutting incident response time by approximately 40%.
- Created PySpark and BigQuery ETL pipelines processing terabyte-scale logs for near real-time anomaly detection and reliability tracking.
- Productionized pipelines and models using Docker, Kubernetes, and Terraform with Tekton CI/CD and SonarQube checks, accelerating release cycles by 10-15%.
- Delivered executive Looker/Tableau dashboards for security and reliability metrics to support prioritization and remediation tracking.

CloudEarl Solutions — Data Engineer

January 2021 - August 2021

Built cross-cloud ETL and reliability controls to support analytics and downstream ML workflows.

- Designed and automated ingestion and transformation pipelines across AWS, Azure, and GCP, reducing analytics data latency by 20%.
- Built Python monitoring pipelines (boto3/AWS SDK) to collect operational metrics from S3, ECS, Lambda, and CloudWatch and store normalized data in PostgreSQL for reporting and alerts.
- Containerized workflows with Docker and automated deployments via Jenkins, reducing manual releases and speeding deployment cycles.
- Implemented data quality controls (validation, schema enforcement, monitoring alerts), reducing manual data fixes by 25%.

PROJECTS

Emotion Detection System (TensorFlow, PyTorch, OpenCV, AWS SageMaker) — Built a real-time facial emotion recognition system using CNN architectures and data augmentation techniques, achieving 85% classification accuracy. Deployed via SageMaker with GPU inference for low-latency performance.

FinOpsRAG - Cloud Cost Insight Assistant (Python, SQL, AWS S3, Athena, FastAPI, OpenSearch) — Built an AWS cost analysis pipeline ingesting Cost & Usage Report data from S3 and querying via Athena for drilldowns by service, account, and team. Added a RAG layer over billing guidelines and past tickets to explain cost spikes with citations and surface optimization opportunities like idle compute and underutilized resources.

PUBLICATIONS & HONORS

Hyperparameter Tuning for Hyperspectral Image Classification — IEEE Conference, 2021

Contribution Toward Cloud — Ford Motor Company, 2022

EDUCATION

University of Central Missouri — Master of Science, Computer Science, GPA 3.6/4.0 | Lee's Summit, MO | May 2023 – Dec 2024

Centurion University of Technology and Management — Bachelor of Technology, Computer Science, GPA 3.6/4.0 |

Visakhapatnam, India | May 2017 – July 2021